



MRC Cognition
and Brain
Sciences Unit



UNIVERSITY OF
CAMBRIDGE

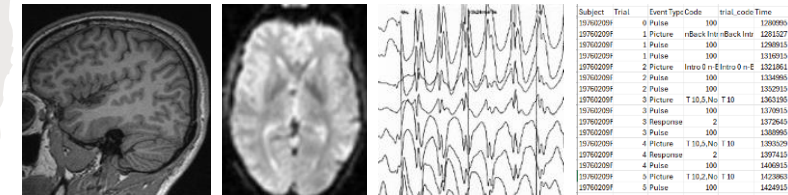
Brain Imaging Data Organisation

Dace Apšvalka
[Datza]

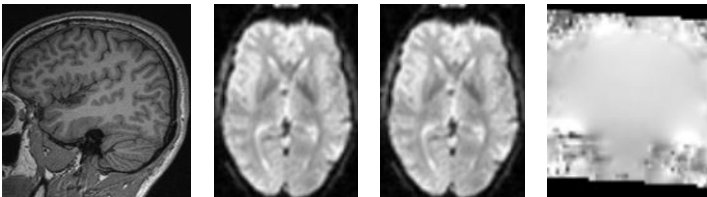
COGNESTIC, 2026

Complexity of Brain Imaging Data

- A single study often involves:
 - Multiple imaging protocols
 - Many participants, sometimes across sessions
- Produces numerous files in diverse formats:
 - From simple text files (logs, metadata)
 - To large multidimensional images
- Datasets become large, diverse, and complex



Complexity of Brain Imaging Data



Subject	Trial	Event Type Code	trial_code	Time	T
157602009	0	Pulse	100	1208995	
157602009	1	Picture	rbBack InrbBack Inrb	12201527	
157602009	1	Pulse	100	1290915	
157602009	1	Pulse	100	1310015	
157602009	2	Picture	Inrb0 nrb Inrb0 nrb	13221861	
157602009	2	Pulse	100	1334895	
157602009	2	Pulse	100	1352915	
157602009	3	Picture	T10.5.No T10	1365195	
157602009	3	Pulse	100	1370915	
157602009	3	Response	2	1372645	
157602009	3	Pulse	100	1380955	
157602009	4	Picture	T10.5.No T10	1393329	
157602009	4	Response	2	1397145	
157602009	4	Pulse	100	1400915	
157602009	5	Picture	T10.2.No T10	1422963	
157602009	5	Pulse	100	1424915	

scans/

```
├─ subj01/
│   ├── s01_t1w.nii.gz
│   ├── s01_bold1_run1.nii.gz
│   ├── s01_bold1_run2.nii.gz
│   └─ s01_fieldmap.nii.gz
├─ subj02/
│   ├── s02_t1w.nii.gz
│   ├── s02_bold1_run1.nii.gz
│   ├── s02_bold1_run2.nii.gz
│   └─ s02_fieldmap.nii.gz
```

...

```
├─ subj100/
│   └─ s100_t1w.nii.gz
└─ ...
```

logs/

```
├─ subj01/
│   ├── events_taskA_run1.tsv
│   ├── events_taskA_run2.tsv
├─ subj02/
│   ├── events_taskA_run1.tsv
│   └─ events_taskA_run2.tsv
```

...

```
├─ subj100/
└─ ...
```

protocols/

...

and even more files
(more modalities, sessions,
tasks)

Complexity of Brain Imaging Data



Subject	Task	Session	Run	File Name	File Size
001	Rest	1	1	001_R1_T1w.nii.gz	1.2 GB
001	Rest	1	2	001_R2_T1w.nii.gz	1.2 GB
001	Rest	1	3	001_R3_T1w.nii.gz	1.2 GB
001	Rest	1	4	001_R4_T1w.nii.gz	1.2 GB
001	Rest	1	5	001_R5_T1w.nii.gz	1.2 GB
001	Rest	1	6	001_R6_T1w.nii.gz	1.2 GB
001	Rest	1	7	001_R7_T1w.nii.gz	1.2 GB
001	Rest	1	8	001_R8_T1w.nii.gz	1.2 GB
001	Rest	1	9	001_R9_T1w.nii.gz	1.2 GB
001	Rest	1	10	001_R10_T1w.nii.gz	1.2 GB
001	Rest	1	11	001_R11_T1w.nii.gz	1.2 GB
001	Rest	1	12	001_R12_T1w.nii.gz	1.2 GB
001	Rest	1	13	001_R13_T1w.nii.gz	1.2 GB
001	Rest	1	14	001_R14_T1w.nii.gz	1.2 GB
001	Rest	1	15	001_R15_T1w.nii.gz	1.2 GB
001	Rest	1	16	001_R16_T1w.nii.gz	1.2 GB
001	Rest	1	17	001_R17_T1w.nii.gz	1.2 GB
001	Rest	1	18	001_R18_T1w.nii.gz	1.2 GB
001	Rest	1	19	001_R19_T1w.nii.gz	1.2 GB
001	Rest	1	20	001_R20_T1w.nii.gz	1.2 GB

Many possible ways to name and organise the data

```
scans/  
├── subj01/  
│   ├── s01_t1w.nii.gz  
│   ├── s01_bold1_run1.nii.gz  
│   ├── s01_bold1_run2.nii.gz  
│   └── s01_fieldmap.nii.gz  
├── subj02/  
│   ├── s02_t1w.nii.gz  
│   ├── s02_bold1_run1.nii.gz  
│   ├── s02_bold1_run2.nii.gz  
│   └── s02_fieldmap.nii.gz  
├── subj100/  
│   ├── s100_t1w.nii.gz  
│   └── s100_bold1_run1.nii.gz  
└── ...  
├── subj01/  
│   ├── events_taskA_run1.tsv  
│   └── events_taskA_run2.tsv  
├── subj02/  
│   ├── events_taskA_run1.tsv  
│   └── events_taskA_run2.tsv  
├── ...  
└── subj100/  
    ├── ...  
    └── ...  
protocols/  
├── ...  
└── ...
```

More files
(more modalities, sessions,
tasks)

Inconsistent Data Organisation

- Difficult for others (and you!) to understand data and track changes
- Scripts must be adapted, can't be easily reused
- Huge effort to automate workflows, no way to automatically validate datasets
- Increased risk of errors (wrong files, outdated versions)
- Time wasted searching and reorganising
- Harder to reproduce results and collaborate



Inconsistent Data Organisation

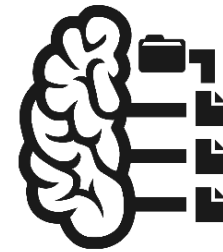


- Difficult for others (and you!) to understand data and track changes
- Scripts must be adapted, can't be easily reused
- Hard effort to automate workflows, no way to automatically validate datasets
- Increased risk of errors (wrong files, outdated versions)
- Time wasted searching and reorganising
- Harder to reproduce results and collaborate

Wouldn't it be much easier if everybody organised the files in the same way?

Brain Imaging Data Structure (BIDS)

- A standard for organising data and metadata across various neuroscience modalities (MRI, MEG, EEG, PET)
- Enables easier sharing, reuse, and application of automated pipelines and quality assurance protocols



BIDS
BRAIN IMAGING DATA STRUCTURE

Brain Imaging Data Structure (BIDS)

- A standard for organising data and metadata across various neuroscience modalities (MRI, MEG, EEG, PET)
- Enables easier sharing, reuse, and application of automated pipelines and quality assurance protocols



SCIENTIFIC DATA

OPEN

SUBJECT CATEGORIES

- » Data publication and archiving
- » Research data

The brain imaging data structure, a format for organizing and describing outputs of neuroimaging experiments

Krzysztof J. Gorgolewski¹, Tibor Auer², Vince D. Calhoun^{3,4}, R. Cameron Craddock^{5,6}, Samir Das⁷, Eugene P. Duff⁸, Guillaume Flandin⁹, Satrajit S. Ghosh^{10,11}, Tristan Glatard^{7,12}, Yaroslav O. Halchenko¹³, Daniel A. Handwerker¹⁴, Michael Hanke^{15,16}, David Keator¹⁷, Xiangrui Li¹⁸, Zachary Michael¹⁹, Camille Maumet²⁰, B. Nolan Nichols^{21,22}, Thomas E. Nichols^{20,23}, John Pellman⁶, Jean-Baptiste Poline²⁴, Ariel Rokem²⁵, Gunnar Schaefer^{1,26}, Vanessa Sochat²⁷, William Triplett¹, Jessica A. Turner^{3,28}, Gaël Varoquaux²⁹ & Russell A. Poldrack¹

Received: 18 December 2015

Accepted: 19 May 2016

Published: 21 June 2016



RESEARCH ARTICLE

BIDS apps: Improving ease of use, accessibility, and reproducibility of neuroimaging data analysis methods

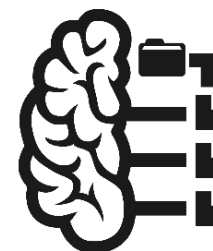
Krzysztof J. Gorgolewski^{1*}, Fidel Alfaro-Almagro², Tibor Auer³, Pierre Bellec^{4,5}, Mihai Capota⁶, M. Mallar Chakravarty^{7,8}, Nathan W. Churchill⁹, Alexander Li Cohen¹⁰, R. Cameron Craddock^{11,12}, Gabriel A. Devenyi^{7,8}, Anders Eklund^{13,14,15}, Oscar Esteban¹, Guillaume Flandin¹⁶, Satrajit S. Ghosh^{17,18}, J. Swaroop Guntupalli¹⁹, Mark Jenkinson², Anisha Keshavan²⁰, Gregory Kiar^{21,22}, Franziskus Liem²³, Pradeep Reddy Raamana^{24,25}, David Raffel²⁶, Christopher J. Steele^{7,8}, Pierre-Olivier Quirion¹⁵, Robert E. Smith²⁶, Stephen C. Strother^{24,25}, Gaël Varoquaux²⁷, Yida Wang⁶, Tal Yarkoni²⁸, Russell A. Poldrack¹




```

sub-15
├── ses-mri
│   ├── anat
│   │   ├── sub-15_ses-mri_T1w.json
│   │   └── sub-15_ses-mri_T1w.nii.gz
│   ├── fmap
│   │   ├── sub-15_ses-mri_acq-func_magnitude1.json
│   │   ├── sub-15_ses-mri_acq-func_magnitude1.nii.gz
│   │   ├── sub-15_ses-mri_acq-func_magnitude2.json
│   │   ├── sub-15_ses-mri_acq-func_magnitude2.nii.gz
│   │   ├── sub-15_ses-mri_acq-func_phasediff.json
│   │   └── sub-15_ses-mri_acq-func_phasediff.nii.gz
│   └── func
│       ├── sub-15_ses-mri_task-facerecognition_run-01_bold.json
│       ├── sub-15_ses-mri_task-facerecognition_run-01_bold.nii.gz
│       ├── sub-15_ses-mri_task-facerecognition_run-01_events.tsv
│       ├── sub-15_ses-mri_task-facerecognition_run-02_bold.json
│       ├── sub-15_ses-mri_task-facerecognition_run-02_bold.nii.gz
│       ├── sub-15_ses-mri_task-facerecognition_run-02_events.tsv
│       ├── sub-15_ses-mri_task-facerecognition_run-03_bold.json
│       ├── sub-15_ses-mri_task-facerecognition_run-03_bold.nii.gz
│       ├── sub-15_ses-mri_task-facerecognition_run-03_events.tsv
│       ├── sub-15_ses-mri_task-facerecognition_run-04_bold.json
│       ├── sub-15_ses-mri_task-facerecognition_run-04_bold.nii.gz
│       ├── sub-15_ses-mri_task-facerecognition_run-04_events.tsv
│       ├── sub-15_ses-mri_task-facerecognition_run-05_bold.json
│       ├── sub-15_ses-mri_task-facerecognition_run-05_bold.nii.gz
│       ├── sub-15_ses-mri_task-facerecognition_run-05_events.tsv
│       ├── sub-15_ses-mri_task-facerecognition_run-06_bold.json
│       ├── sub-15_ses-mri_task-facerecognition_run-06_bold.nii.gz
│       ├── sub-15_ses-mri_task-facerecognition_run-06_events.tsv
│       ├── sub-15_ses-mri_task-facerecognition_run-07_bold.json
│       ├── sub-15_ses-mri_task-facerecognition_run-07_bold.nii.gz
│       ├── sub-15_ses-mri_task-facerecognition_run-07_events.tsv
│       ├── sub-15_ses-mri_task-facerecognition_run-08_bold.json
│       ├── sub-15_ses-mri_task-facerecognition_run-08_bold.nii.gz
│       ├── sub-15_ses-mri_task-facerecognition_run-08_events.tsv
│       ├── sub-15_ses-mri_task-facerecognition_run-09_bold.json
│       ├── sub-15_ses-mri_task-facerecognition_run-09_bold.nii.gz
│       └── sub-15_ses-mri_task-facerecognition_run-09_events.tsv
└── sub-15_ses-mri_scans.tsv

```



BIDS
BRAIN IMAGING DATA STRUCTURE

Today's plan



MRI Data Organisation



MRI Data Manipulation

```

[[ [ 0., 0., 0., ..., 0., 0., 0.],
  [ 0., 0., 0., ..., 0., 0., 0.],
  [ 0., 0., 0., ..., 0., 0., 0.],
  ...,
  [ 0., 0., 0., ..., 0., 0., 0.],
  [ 0., 0., 0., ..., 0., 0., 0.],
  [ 0., 0., 0., ..., 0., 0., 0.]],

[[ [ 0., 0., 0., ..., 0., 0., 0.],
  [ 0., 25., 23., ..., 23., 32., 0.],
  [ 0., 28., 21., ..., 25., 25., 0.],
  ...,
  [ 0., 26., 24., ..., 40., 20., 0.],
  [ 0., 44., 28., ..., 30., 21., 0.],
  [ 0., 0., 0., ..., 0., 0., 0.]],

[[ [ 0., 0., 0., ..., 0., 0., 0.],
  [ 0., 28., 26., ..., 31., 29., 0.],
  [ 0., 32., 30., ..., 22., 21., 0.],
  ...,
  [ 0., 27., 24., ..., 31., 30., 0.],
  [ 0., 30., 23., ..., 37., 22., 0.],
  [ 0., 0., 0., ..., 0., 0., 0.]],

....

```